

A thick dark blue vertical bar is positioned on the left side of the page. To its right, several thin, curved lines in shades of blue and grey sweep upwards and to the right, creating a dynamic, abstract design.

**SWT 22022 – Practical
for Internet Application
Development**

Department of Information
and Communication
Technology
Faculty of Technology

**Lab sheet :12
Reg. Number: SEU/IS/20/ICT/084
Academic Year :2020/2021
Date: 2024.10.02
Practical No :12**

Title: Introduction to PHP framework (Laravel)

Aims:

- Setting up development environment
- Database connectivity
- Project creation
- Understanding the Laravel's core concepts
- Project structure navigation

Tasks:

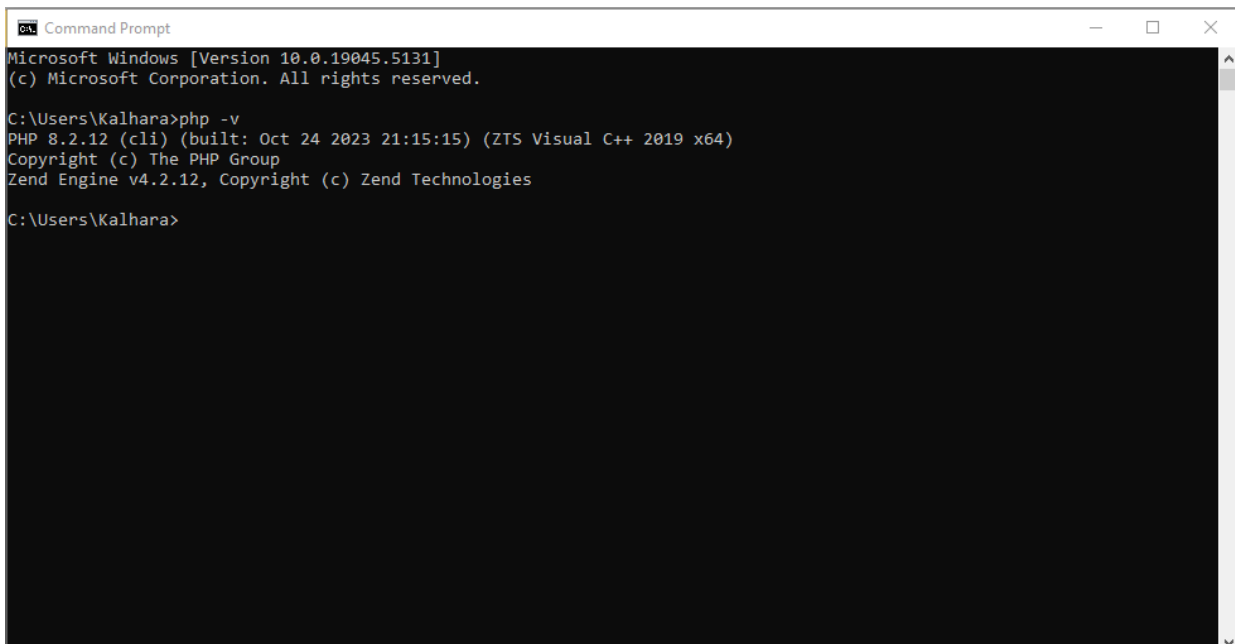
- Install PHP
- Install Laravel and Composer
- Create a Laravel project
- Navigate through Laravel documentation and architecture
- Introduce Laravel database
- Review of Laravel file structure

Activities :

1. Install PHP

On Linux and macOS, PHP is typically pre-installed. To check the version of PHP installed, run the following command:

php -v



```
Command Prompt
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Kalhara>php -v
PHP 8.2.12 (cli) (built: Oct 24 2023 21:15:15) (ZTS Visual C++ 2019 x64)
Copyright (c) The PHP Group
Zend Engine v4.2.12, Copyright (c) Zend Technologies

C:\Users\Kalhara>
```

If PHP is not installed, you can install it using your package manager.

On *Ubuntu*, you can install PHP with the following command:

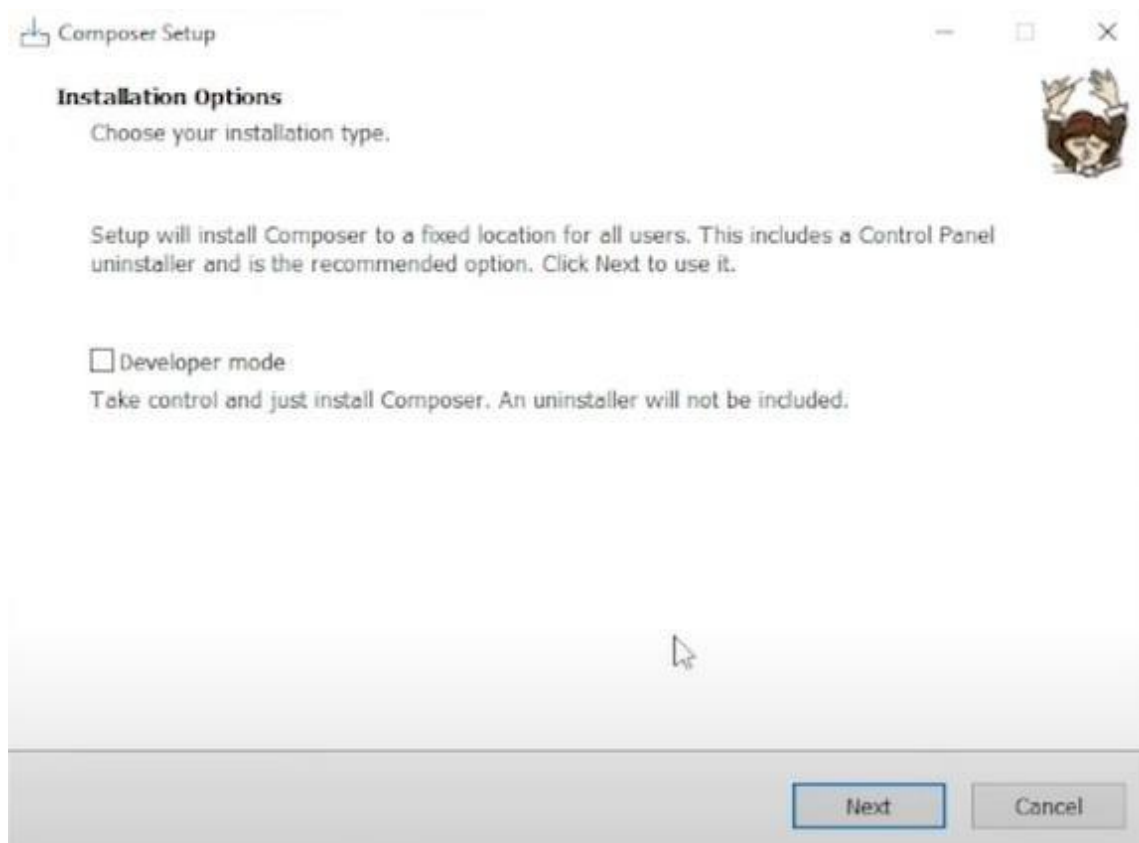
sudo apt install php

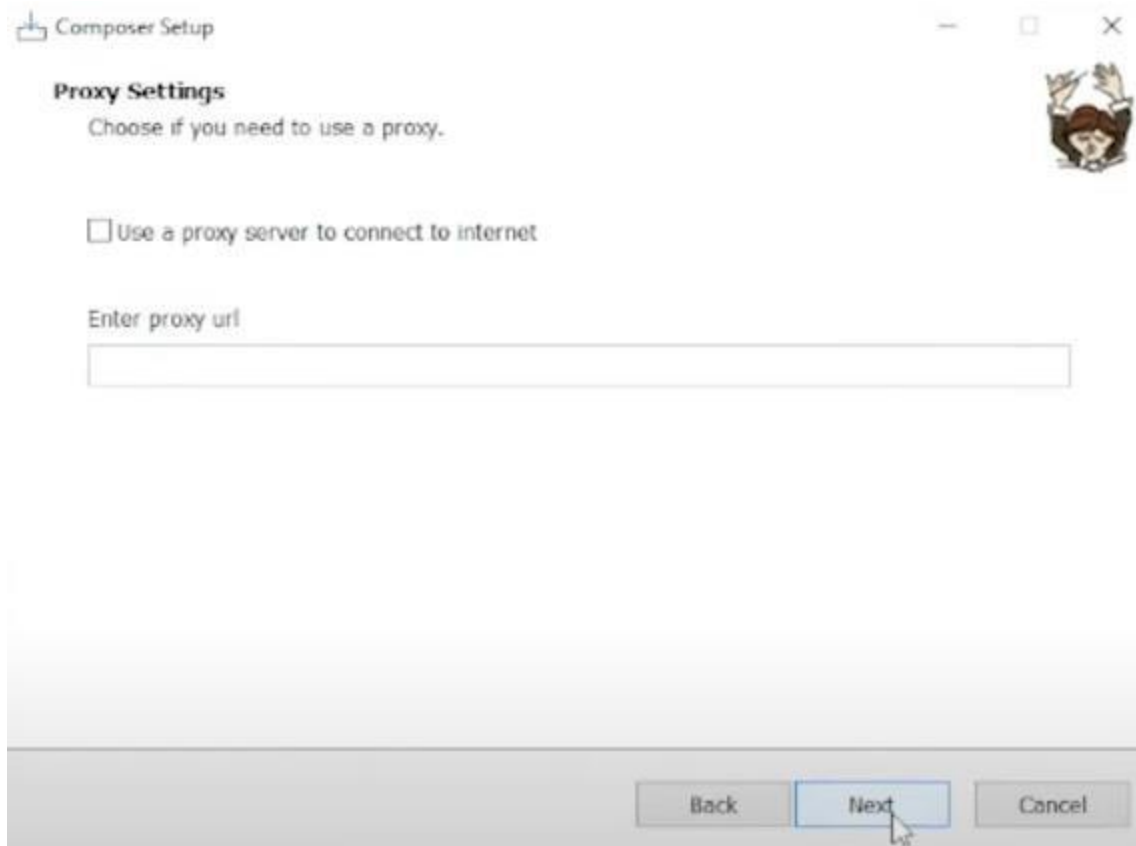
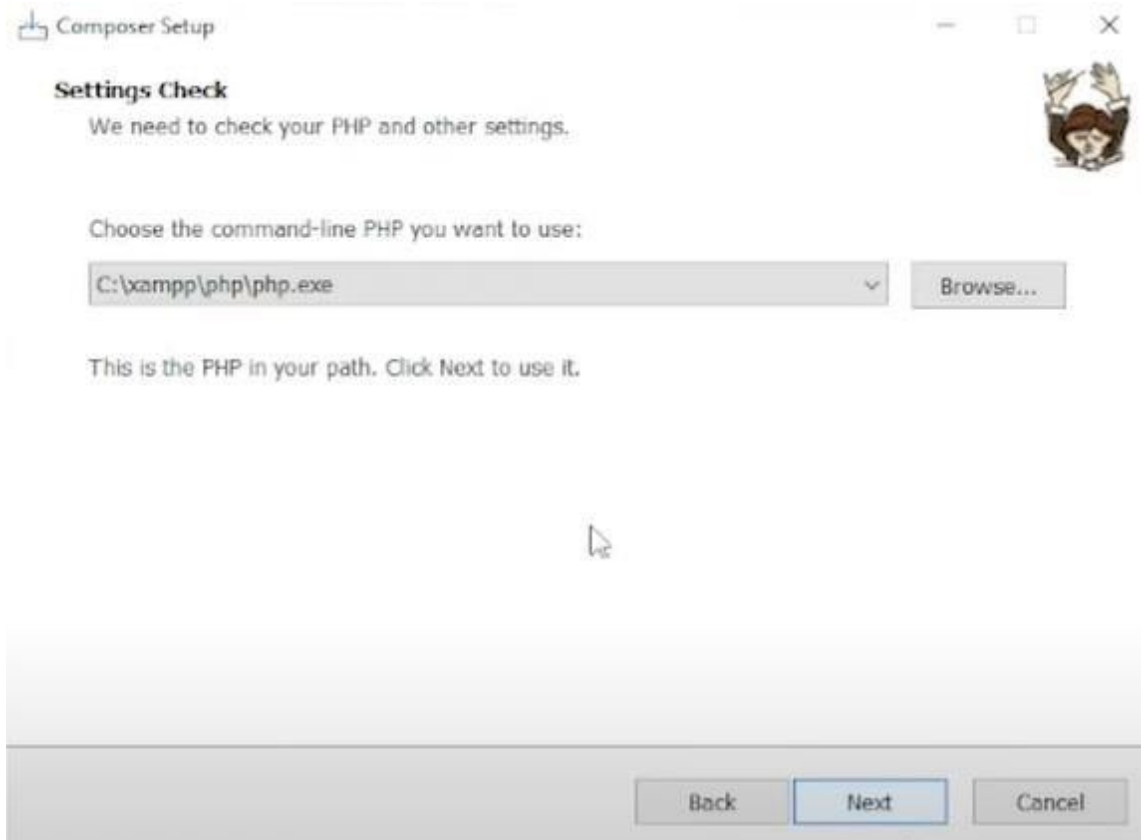
On *Windows*, You can refer this documentation :

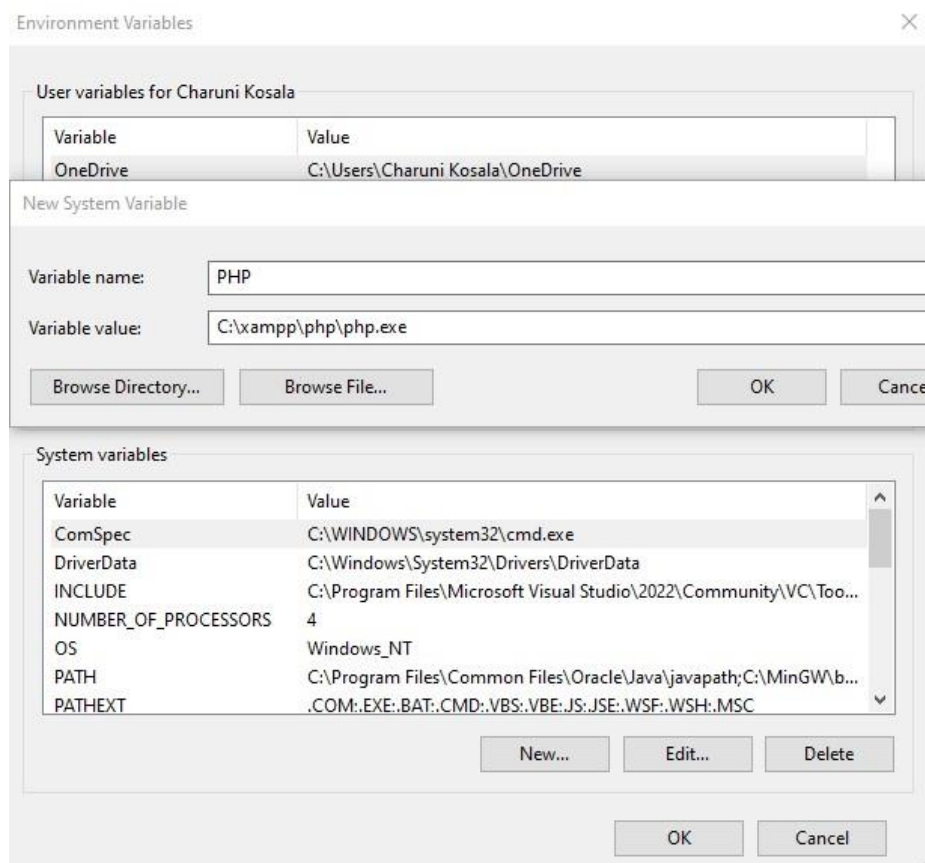
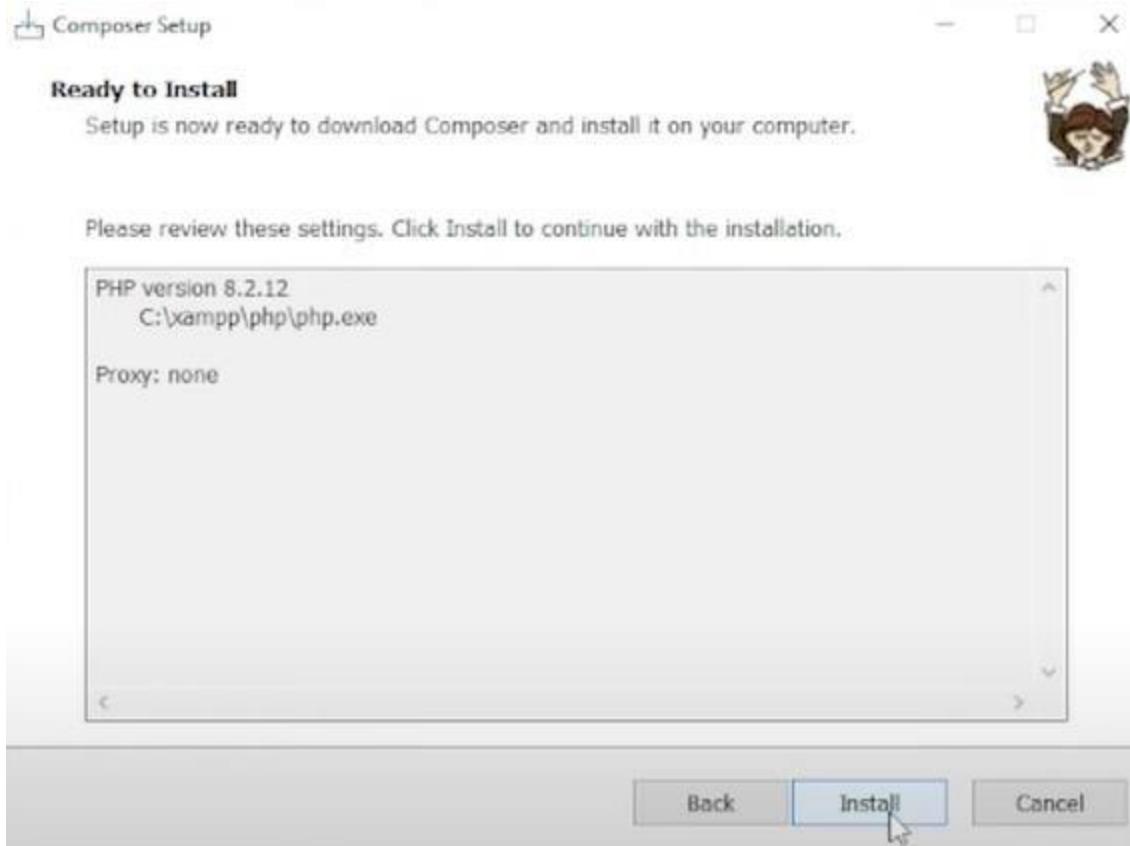
<https://windows.php.net/download>

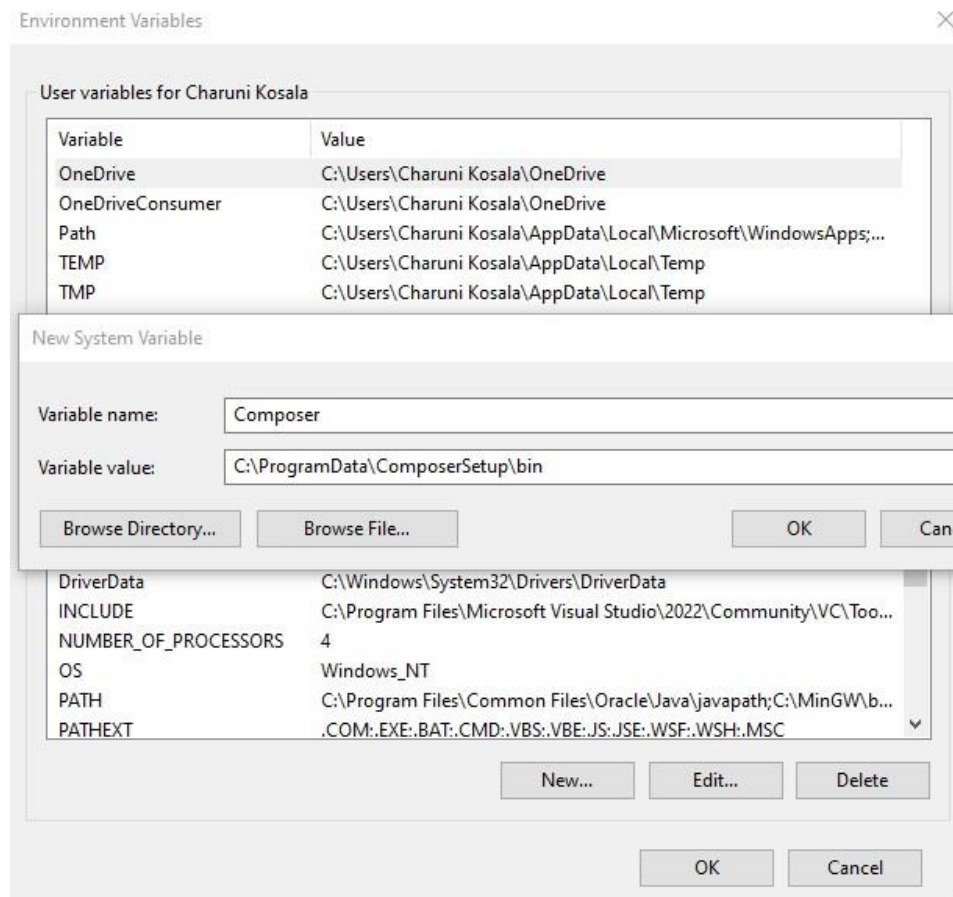
2. Install Laravel and Composer

<https://getcomposer.org/doc/00-intro.md>









. To check the version of PHP installed, run the following command:

composer –version

```

C:\Users\Kaiharax\composer -v

Composer version 2.7.5 2024-09-04 14:43:28

Usage:
  command [options] [arguments]

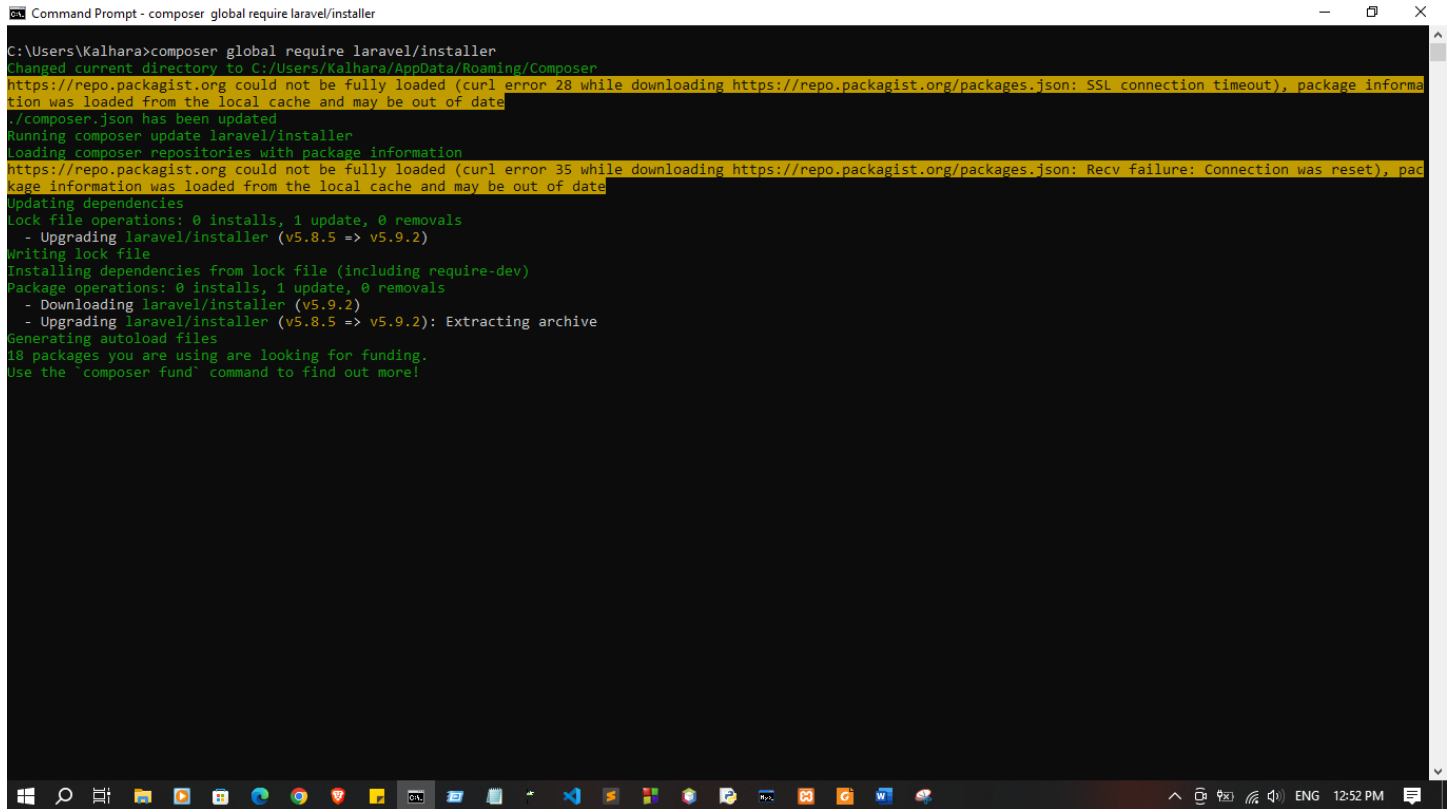
Options:
  -h, --help                Display help for the given command. When no command is given display help for the list command
  -q, --quiet               Do not output any message
  -V, --version             Display this application version
  --ansi|--no-ansi          Force (or disable --no-ansi) ANSI output
  -n, --no-interaction      Do not ask any interactive question
  --profile                Display timing and memory usage information
  --no-plugins              Whether to disable plugins.
  --no-scripts              Skips the execution of all scripts defined in composer.json file.
  -d, --working-dir=WORKING-DIR If specified, use the given directory as working directory.
  --no-cache                Prevent use of the cache
  -v|vv|vvv, --verbose     Increase the verbosity of messages: 1 for normal output, 2 for more verbose output and 3 for debug

Available commands:
  about                Shows a short information about Composer
  archive              Creates an archive of this composer package
  audit                Checks for security vulnerability advisories for installed packages
  browse              [home] Opens the package's repository URL or homepage in your browser
  dump                Increases the lower limit of your composer.json requirements to the currently installed versions
  check-platform-reqs Check that platform requirements are satisfied
  clear-cache          [clear-cache(cc)] Clears composer's internal package cache
  completion           Dump the shell completion script
  config              Sets config options
  create-project        Creates new project from a package into given directory
  depends              [why] Shows which packages cause the given package to be installed
  diagnose             Diagnoses the system to identify common errors
  dump-autoload        [dumpautoload] Dumps the autoloader
  exec                 Executes a vendored binary/script
  fund                Discover how to help fund the maintenance of your dependencies
  global               Allows running commands in the global composer dir ($COMPOSER_HOME)
  help                Display help for a command
  
```

3. Install Laravel

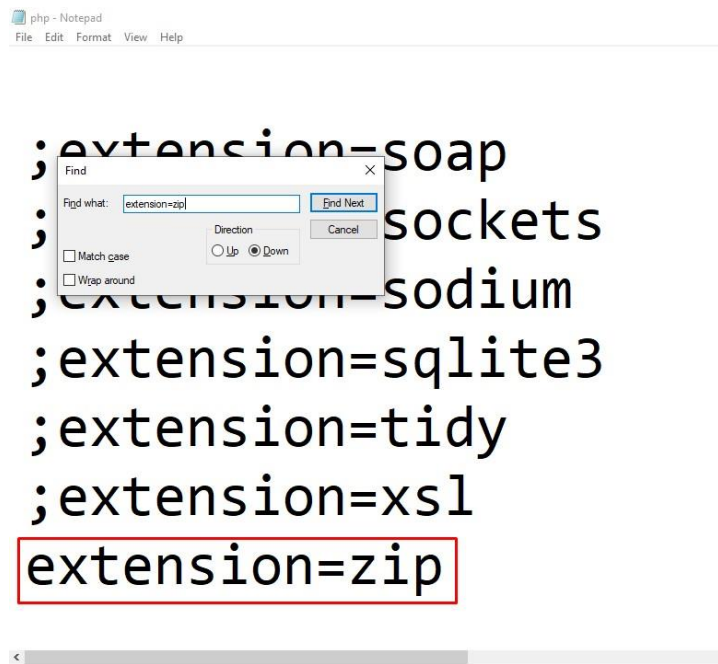
To install Laravel, run the following command:

composer global require laravel/installer



```
C:\Users\Kalhara>composer global require laravel/installer
Changed current directory to C:/Users/Kalhara/AppData/Roaming/Composer
https://repo.packagist.org could not be fully loaded (curl error 28 while downloading https://repo.packagist.org/packages.json: SSL connection timeout), package information was loaded from the local cache and may be out of date
./composer.json has been updated
Running composer update laravel/installer
Loading composer repositories with package information
https://repo.packagist.org could not be fully loaded (curl error 35 while downloading https://repo.packagist.org/packages.json: Recv failure: Connection was reset), package information was loaded from the local cache and may be out of date
Updating dependencies
Lock file operations: 0 installs, 1 update, 0 removals
- Upgrading laravel/installer (v5.8.5 => v5.9.2)
Writing lock file
Installing dependencies from lock file (including require-dev)
Package operations: 0 installs, 1 update, 0 removals
- Downloading laravel/installer (v5.9.2)
- Upgrading laravel/installer (v5.8.5 => v5.9.2): Extracting archive
Generating autoload files
18 packages you are using are looking for funding.
Use the "composer fund" command to find out more!
```

Do these changes in php.ini file



php - Notepad
File Edit Format View Help

```
; See https://github.com/php/extension-ldap  
extension-ldap  
extension-curl  
;extension=ffi  
;extension=ftp  
extension=fileinfo  
extension=gd  
extension=gettext
```

Find
Find what: extension=file
Find Next
Cancel
Direction
☒ Up ☐ Down
☐ Match case
☐ Wrap around

php - Notepad
File Edit Format View Help

```
;extension=intl  
;extension=imap  
extension=mbstring  
extension=exif ; M  
extension=mysqli  
;extension=oci8_12c ;  
;extension=oci8_19 ; U  
;extension=odbc  
;extension=openssl
```

Find
Find what: extension=mysql
Find Next
Cancel
Direction
☐ Up ☒ Down
☐ Match case
☐ Wrap around

php - Notepad
File Edit Format View Help

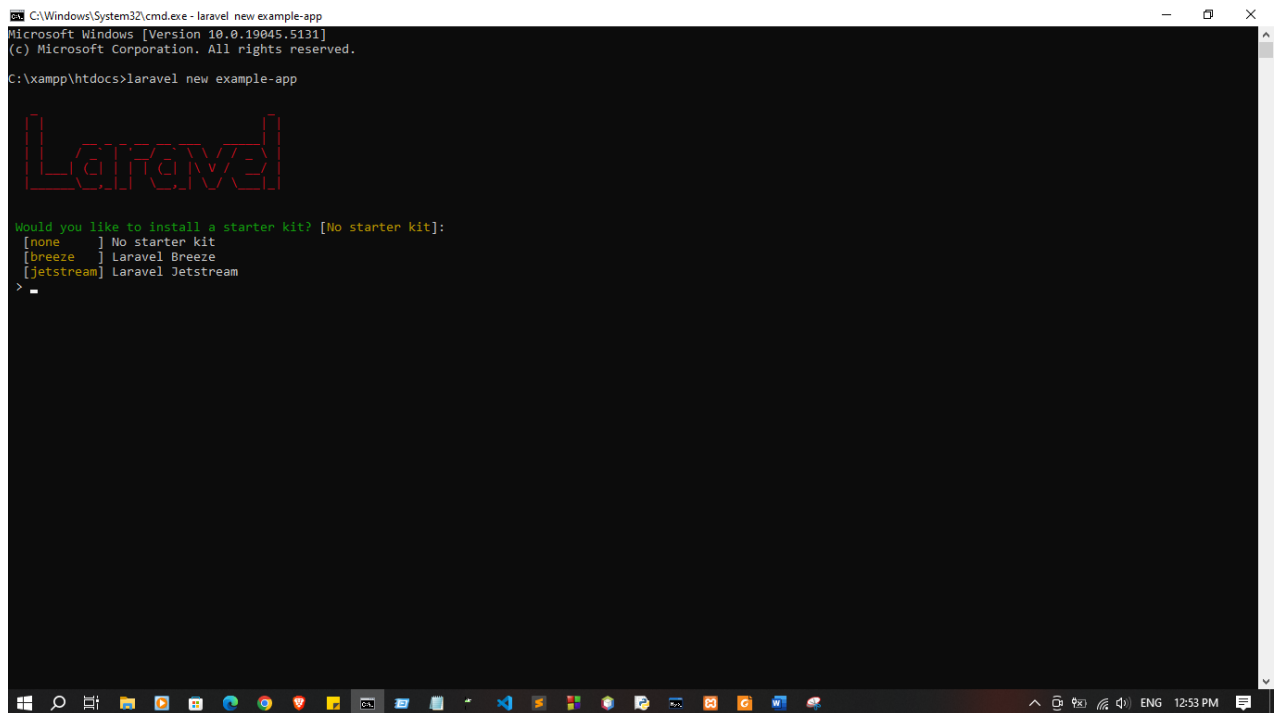
```
;extension=imap  
extension=mbstring  
extension=exif ; Mus  
extension=mysqli  
;extension=oci8_12c ; Us  
;extension=oci8_19 ; Use  
;extension=odbc  
;extension=openssl  
;extension=pdo_firebird
```

Find
Find what: extension=openssl
Find Next
Cancel
Direction
☐ Up ☒ Down
☐ Match case
☐ Wrap around

4. Create Laravel project

Use this command :

laravel new example-app

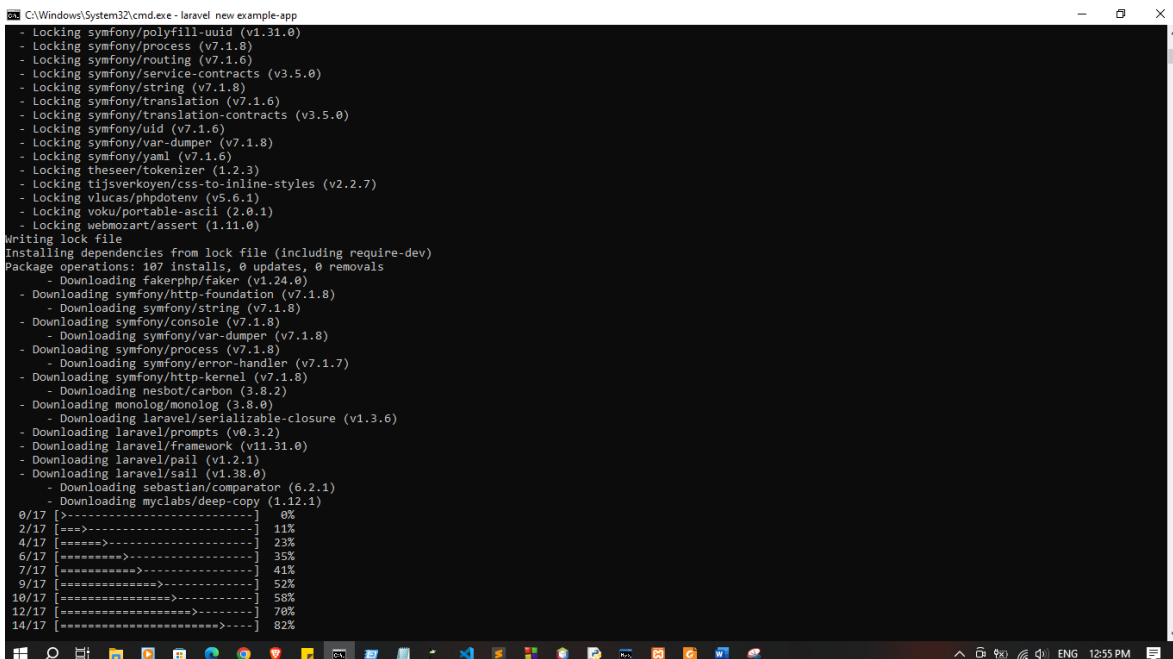


```
C:\Windows\System32\cmd.exe - laravel new example-app
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\xampp\htdocs>laravel new example-app

Laravel

Would you like to install a starter kit? [No starter kit]:
[none] No starter kit
[breeze] Laravel Breeze
[jetstream] Laravel Jetstream
> -
```



```
C:\Windows\System32\cmd.exe - laravel new example-app
- Locking symfony/polyfill-uuid (v1.31.0)
- Locking symfony/process (v7.1.8)
- Locking symfony/routing (v7.1.6)
- Locking symfony/service-contracts (v3.5.0)
- Locking symfony/string (v7.1.8)
- Locking symfony/translation (v7.1.6)
- Locking symfony/translation-contracts (v3.5.0)
- Locking symfony/uid (v7.1.6)
- Locking symfony/var-dumper (v7.1.8)
- Locking symfony/yaml (v7.1.6)
- Locking theseer/tokenizer (1.2.3)
- Locking tijsverkoyen/css-to-inline-styles (v2.2.7)
- Locking vlucas/phpdotenv (v5.6.1)
- Locking voku/portable-ascii (2.0.1)
- Locking webmozart/assert (1.11.0)
Writing lock file
Installing dependencies from lock file (including require-dev)
Package operations: 107 installs, 0 updates, 0 removals
- Downloading fakerphp/faker (v1.24.0)
- Downloading symfony/http-foundation (v7.1.8)
- Downloading symfony/string (v7.1.8)
- Downloading symfony/console (v7.1.8)
- Downloading symfony/var-dumper (v7.1.8)
- Downloading symfony/process (v7.1.8)
- Downloading symfony/error-handler (v7.1.7)
- Downloading symfony/http-kernel (v7.1.8)
- Downloading nesbot/carbon (3.8.2)
- Downloading monolog/monolog (3.8.0)
- Downloading laravel/serializable-closure (v1.3.6)
- Downloading laravel/prompts (v0.3.2)
- Downloading laravel/framework (v11.31.0)
- Downloading laravel/pail (v1.2.1)
- Downloading laravel/sail (v1.38.0)
- Downloading sebastian/comparator (5.2.1)
- Downloading myclabs/deep-copy (1.12.1)
0/17 [>-----] 0%
2/17 [===>-----] 11%
4/17 [=====>-----] 23%
6/17 [=====>-----] 35%
7/17 [=====>-----] 41%
9/17 [=====>-----] 52%
10/17 [=====>-----] 58%
12/17 [=====>-----] 70%
14/17 [=====>-----] 82%
```

```
C:\Windows\System32\cmd.exe
> tsc && vite build

vite v5.4.11 building for production...
transforming...
  352 modules transformed.
rendering chunks...
computing gzip size...
public/build/manifest.json          7.20 kB | gzip: 0.89 kB
public/build/assets/app-DE712row.css 33.75 kB | gzip: 6.48 kB
public/build/assets/InputLabel-wRfDwget.js 0.21 kB | gzip: 0.18 kB
public/build/assets/GuestLayout-coD4_PHo.js 0.50 kB | gzip: 0.33 kB
public/build/assets/PrimaryButton-BS3n5nwu.js 0.50 kB | gzip: 0.34 kB
public/build/assets/TextInput-BUK7m-Z5.js 0.60 kB | gzip: 0.39 kB
public/build/assets/Dashboard-6t0eLWqQ.js 0.64 kB | gzip: 0.37 kB
public/build/assets/Edit-yKaQeBeI.js 1.12 kB | gzip: 0.53 kB
public/build/assets/ForgotPassword-DSze9XJd.js 1.14 kB | gzip: 0.65 kB
public/build/assets/ConfirmPassword-6kzVC3fg.js 1.16 kB | gzip: 0.64 kB
public/build/assets/VerifyEmail-Te-LgC2N.js 1.26 kB | gzip: 0.70 kB
public/build/assets/ResetPassword-sPtqRysY.js 1.76 kB | gzip: 0.69 kB
public/build/assets/Login-DYAdXQOu.js 2.09 kB | gzip: 0.94 kB
public/build/assets/Register-CLH6cSqX.js 2.28 kB | gzip: 0.82 kB
public/build/assets/UpdateProfileInformationForm-9nK2sTTD.js 2.29 kB | gzip: 1.00 kB
public/build/assets/UpdatePasswordForm-BXuWk4KP.js 2.38 kB | gzip: 0.89 kB
public/build/assets/ApplicationLogo-C2z023pR.js 3.12 kB | gzip: 1.33 kB
public/build/assets/AuthenticatedLayout-C0eldIk.js 5.87 kB | gzip: 1.90 kB
public/build/assets/transition-DKGVCN1S.js 14.60 kB | gzip: 5.79 kB
public/build/assets/Welcome-BULi6V2G.js 19.29 kB | gzip: 5.89 kB
public/build/assets/DeleteUserForm-DnCXXU2W.js 29.93 kB | gzip: 10.63 kB
public/build/assets/app-BvIYOSJA.js 257.66 kB | gzip: 86.38 kB
  built in 5.62s

INFO Breeze scaffolding installed successfully.

[INFO] Application ready in [example-app]. You can start your local development using:

  cd example-app
  npm install && npm run build
  composer run dev

New to Laravel? Check out our bootcamp and documentation. Build something amazing!

C:\xampp\htdocs>
```

Run the project :

cd example-app

```
C:\Windows\System32\cmd.exe

vite v5.4.11 building for production...
transforming...
  352 modules transformed.
rendering chunks...
computing gzip size...
public/build/manifest.json          7.20 kB | gzip: 0.89 kB
public/build/assets/app-DE712row.css 33.75 kB | gzip: 6.48 kB
public/build/assets/InputLabel-wRfDwget.js 0.21 kB | gzip: 0.18 kB
public/build/assets/GuestLayout-coD4_PHo.js 0.50 kB | gzip: 0.33 kB
public/build/assets/PrimaryButton-BS3n5nwu.js 0.50 kB | gzip: 0.34 kB
public/build/assets/TextInput-BUK7m-Z5.js 0.60 kB | gzip: 0.39 kB
public/build/assets/Dashboard-6t0eLWqQ.js 0.64 kB | gzip: 0.37 kB
public/build/assets/Edit-yKaQeBeI.js 1.12 kB | gzip: 0.53 kB
public/build/assets/ForgotPassword-DSze9XJd.js 1.14 kB | gzip: 0.65 kB
public/build/assets/ConfirmPassword-6kzVC3fg.js 1.16 kB | gzip: 0.64 kB
public/build/assets/VerifyEmail-Te-LgC2N.js 1.26 kB | gzip: 0.70 kB
public/build/assets/ResetPassword-sPtqRysY.js 1.76 kB | gzip: 0.69 kB
public/build/assets/Login-DYAdXQOu.js 2.09 kB | gzip: 0.94 kB
public/build/assets/Register-CLH6cSqX.js 2.28 kB | gzip: 0.82 kB
public/build/assets/UpdateProfileInformationForm-9nK2sTTD.js 2.29 kB | gzip: 1.00 kB
public/build/assets/UpdatePasswordForm-BXuWk4KP.js 2.38 kB | gzip: 0.89 kB
public/build/assets/ApplicationLogo-C2z023pR.js 3.12 kB | gzip: 1.33 kB
public/build/assets/AuthenticatedLayout-C0eldIk.js 5.87 kB | gzip: 1.90 kB
public/build/assets/transition-DKGVCN1S.js 14.60 kB | gzip: 5.79 kB
public/build/assets/Welcome-BULi6V2G.js 19.29 kB | gzip: 5.89 kB
public/build/assets/DeleteUserForm-DnCXXU2W.js 29.93 kB | gzip: 10.63 kB
public/build/assets/app-BvIYOSJA.js 257.66 kB | gzip: 86.38 kB
  built in 5.62s

INFO Breeze scaffolding installed successfully.

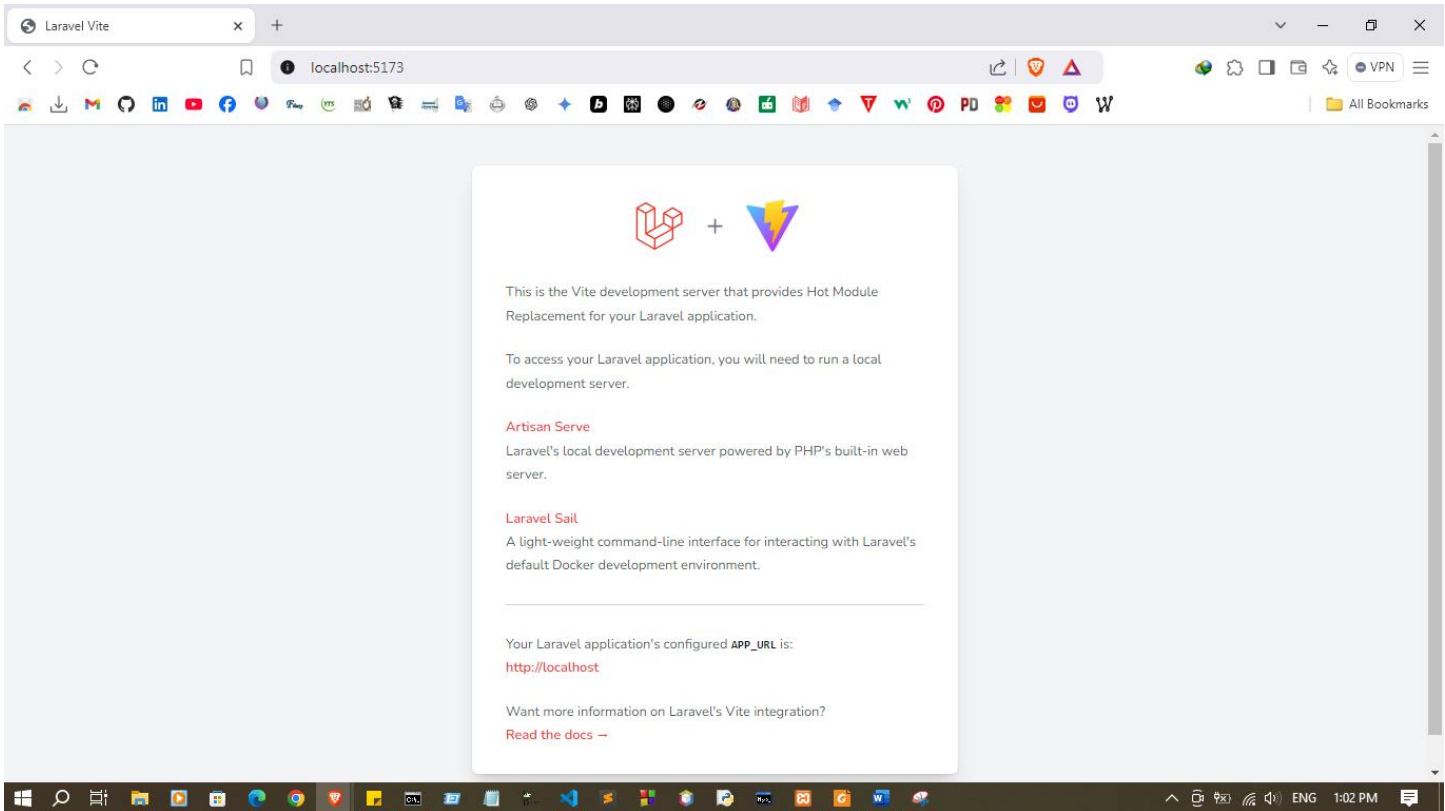
[INFO] Application ready in [example-app]. You can start your local development using:

  cd example-app
  npm install && npm run build
  composer run dev

New to Laravel? Check out our bootcamp and documentation. Build something amazing!

C:\xampp\htdocs>cd example-app
C:\xampp\htdocs\example-app>
```

php artisan serve



5. Navigate through Laravel documentation and architecture

The Laravel documentation is a comprehensive resource for learning about all aspects of the Laravel framework. It can be found at the following URL:

<https://laravel.com/docs/11.x>

The documentation is divided into several sections, including:

- *Getting Started*: This section covers the basics of Laravel, such as setting up a new project, routing, and controllers.
- *Views*: This section covers how to create and render views in Laravel.
- *Blade*: This section covers the Blade templating engine, which is used to create views in Laravel.
- *Artisan*: This section covers the Artisan console, which provides a number of helpful tools for managing Laravel applications.
- *Eloquent*: This section covers Eloquent, the Laravel ORM, which makes it easy to interact with databases.
- *Authentication and Authorization*: This section covers how to implement authentication and authorization in Laravel applications.
- *Testing*: This section covers how to write and run tests for Laravel applications.
- *Deployment*: This section covers how to deploy Laravel applications to production servers.

Architecture:

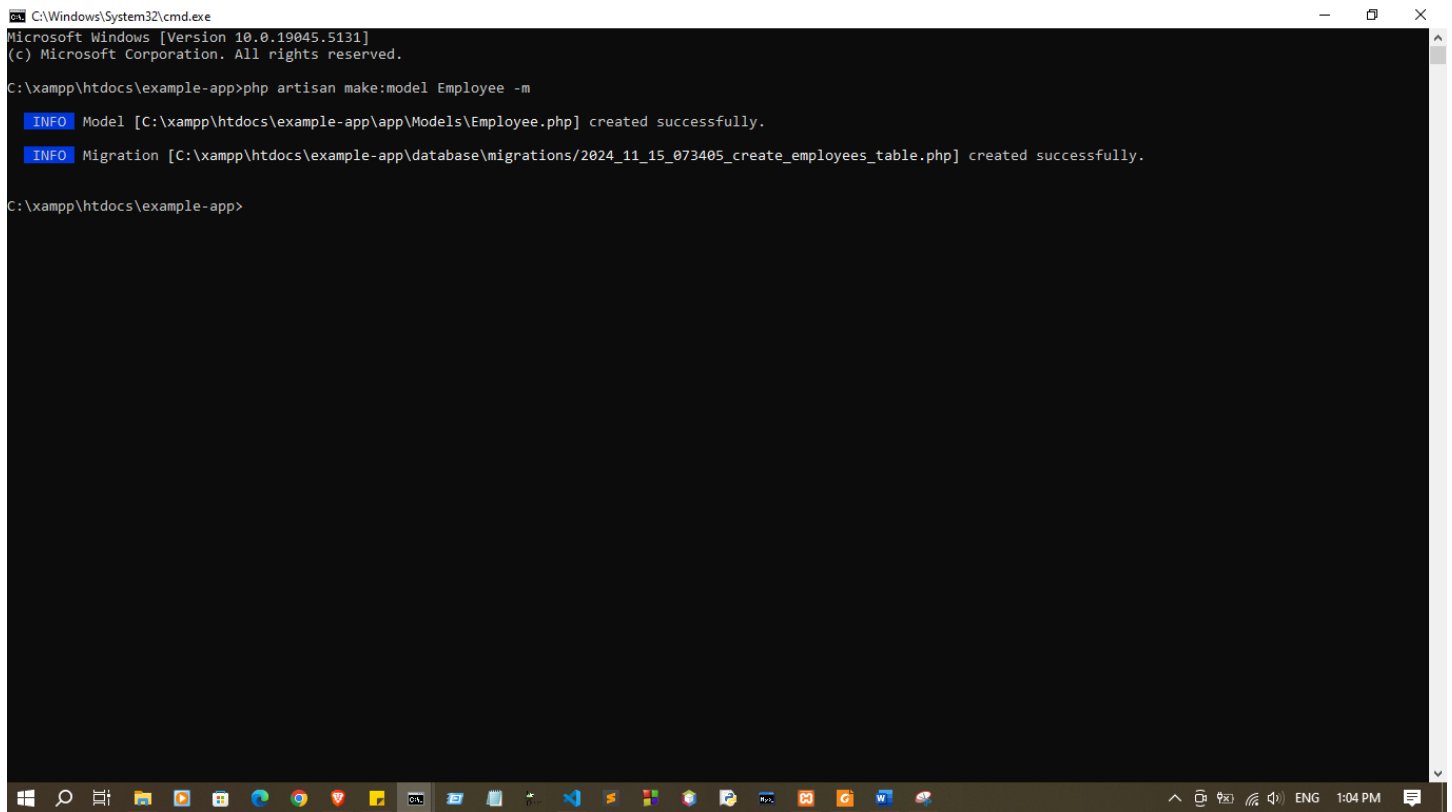
Laravel is a full-stack web framework, meaning that it provides all of the tools needed to develop web applications including:

- *MVC architecture*: Laravel uses the MVC architecture, which separates the application into three layers: models views, and controllers. This makes the code more modular and easier to maintain.
- *Routing*: Laravel provides a robust routing system that allows you to define how your application responds to different HTTP requests.
- *Templating*: Laravel uses the Blade templating engine, which is a powerful and flexible way to create views.
- *Database abstraction layer*: Laravel provides Eloquent, the Laravel ORM, which makes it easy to interact with databases.
- *Authentication and authorization*: Laravel provides built-in support for authentication and authorization, making it easy to secure your applications.
- *Testing framework*: Laravel provides a built-in testing framework that makes it easy to write and run tests for your applications.

6. Introduce Laravel Database

<https://laravel.com/docs/11.x/eloquent>

php artisan make:model Employee -m



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\xampp\htdocs\example-app>php artisan make:model Employee -m

[INFO] Model [C:\xampp\htdocs\example-app\app\Models\Employee.php] created successfully.
[INFO] Migration [C:\xampp\htdocs\example-app\database\migrations\2024_11_15_073405_create_employees_table.php] created successfully.

C:\xampp\htdocs\example-app>
```

```

public function up()
{
    Schema::create('employees', function (Blueprint $table) {
        $table->id();
        $table->string('first_name');
        $table->string('last_name');
        $table->string('email')->unique();
        $table->string('phone')->nullable();
        $table->string('position');
        $table->decimal('salary', 8, 2);
        $table->date('hired_date');
        $table->timestamps();
    });
}

```

```

<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up()
    {
        Schema::create('employees', function (Blueprint $table) {
            $table->id();
            $table->string('first_name');
            $table->string('last_name');
            $table->string('email')->unique();
            $table->string('phone')->nullable();
            $table->string('position');
            $table->decimal('salary', 8, 2);
            $table->date('hired_date');
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('employees');
    }
}

```

```
}  
};
```

```
database > migrations > 2024_11_15_073405_create_employees_table.php > class > up  
1  <?php  
2  
3  use Illuminate\Database\Migrations\Migration;  
4  use Illuminate\Database\Schema\Blueprint;  
5  use Illuminate\Support\Facades\Schema;  
6  
7  return new class extends Migration  
8  {  
9      /**  
10       * Run the migrations.  
11       */  
12     public function up()  
13     {  
14         Schema::create('employees', function (Blueprint $table) {  
15             $table->id();  
16             $table->string('first_name');  
17             $table->string('last_name');  
18             $table->string('email')->unique();  
19             $table->string('phone')->nullable();  
20             $table->string('position');  
21             $table->decimal('salary', 8, 2);  
22             $table->date('hired_date');  
23             $table->timestamps();  
24         });  
25     }  
26  
27     /**  
28      * Reverse the migrations.  
29      */  
30     public function down(): void  
31     {  
32         Schema::dropIfExists('employees');  
33     }  
34 }
```

Discussion:

- This lab practical session focuses on gaining a foundational understanding of Laravel, a widely used PHP framework. The session begins with setting up the development environment by installing PHP, Composer, and Laravel. Students will create a Laravel project to explore its core concepts, such as routing, controllers, and middleware, while also familiarizing themselves with the framework's organized project structure. Emphasis will be placed on understanding Laravel's database connectivity and how it simplifies interaction with databases using its built-in ORM, Eloquent. By navigating the Laravel documentation and exploring the file structure, participants will develop a hands-on appreciation of the framework's architecture, preparing them for more advanced tasks in future projects